

Name: Elijah Henderson, Odion Odion-Davis, Kordell Armstong

Date: 1/22/2025 **Class:** 7

MyDesign Portfolio

Element A Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The details below are commonly found in problem definitions, but the specific assignment given by your instructor may dictate different details. If you make a copy of this template to edit for your own problem definition, be sure to delete the instructions.

Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

Explain your problem in detail, including details that you might consider common knowledge. Use data and statistics to justify your problem statement. Do not provide a bulleted list of facts; rather, organize the problem details in paragraph form. Address as many of the 5 Ws +1 H (who, what, where, when, why, how) as is possible, for example:

- 1) Who is involved in this problem?
- 2) What are they doing?
- 3) What unique situations does the problem occur within, if any?
- 4) What is the size of the issue?
- 5) Where is the problem happening?
- 6) When does the problem occur?
- 7) Why does the problem happen?
- 8) How often does the problem happen?

In order to earn a score of 5, you will need to cite at least five sources that are from the last ten years and that come from 3 or more of the following credible categories: peer reviewed journals, professional organizations, government agencies, local and national newspapers. Use in-line citations to explicitly state which information is from which source. Do not just list all of the citations at the end of the submission or in footnotes.

Name: Elijah Henderson, Odion Odion-Davis, Kordell Armstong

Date: 1/22/2025 **Class:** 7

Stakeholder Concerns and Connections (A3, A4)

Directly indicate at least five primary stakeholders or stakeholder groups. These are the people most directly impacted by the problem. Highlight the concerns of your primary stakeholders and use that to justify why there is a need to address the problem. Consider the breadth of the problem and how it might have social and environmental impacts on secondary stakeholders, which are people who may be impacted by a solution to the problem and are still part of the system.

Problem Importance (A6)

End Element A with a compelling argument for why this problem is important and needs to be solved now. Support this argument with data and statistics.

Problem:

The drink machine needs more syrup. At our school we have a drink machine that provides us with numerous drinks that we can have. Although the drink machine can be a bit weird. Sometimes the drink machine starts running out of syrup and starts giving water instead of giving them the drink they had wanted before. At first we believed this problem was caused by the amount of syrup the machine had and how much was being lost but after talking with Mr. Jones we realized there may be a different problem. Mr. Jones talked to us about how students would waste the drink by simply filling their cups up, taking sips, then just pouring it out just because they didn't like it. He said he feels annoyed when he sees kids do that. We also talked with Mr. Gloster and he said that he feels when he sees the drink machine give water rather than their drink he feels annoyed and he wants to find a way to fix it but he said he doesn't know how to engineer through that. Then we talked with many students and they all had the same opinions. The students we talked with were Malcolm, Cole, Malik, Seth, and Niyah. All of these people said they feel upset when they go to the drink machine to get something like Ice Tea and end up getting water instead. We need to make something that can either increase the amount of syrup in the drink machine or make it to where students don't overuse the drink machine because the people who have the biggest effect on this are those with 3rd lunch. There are three lunch periods and the third lunch is the last one with normally the most problems with the machine because of how much people use it in the 1st and 2nd lunches. The sources we used as evidence and guidance to help us were [Part Towns](#), [Donastar](#), [FBD Frozen](#), [Carbonation System](#), and [Tetrapak](#). They all gave us tips and reasons why the drink machine could be acting like this.

Name: Elijah Henderson, Odion Odion-Davis, Kordell Armstong

Date: 1/22/2025 **Class:** 7

They talk about how the overuse of the machine causes things like this to happen. That's why we need to make a way where the syrup is increased and students don't waste the drinks they have for no good reason.

Stakeholder Concerns and Connection:

Our two main stakeholders are Mr. Gloster and Mr. Jones who both had complaints about the drink machine. Mr. Gloster told us his feelings towards the drink machine and how annoyed he felt when see it run out of his drink he wanted and turn into water. Mr. Jones rather shared that students are the main suspects of the problem and need to learn not to take so much syrup out of the machine by wasting a lot of the drink. Then some other Stakeholders we have students who have all complained that when their favorite drink turns to water they get annoyed and start getting upset. This impacts the third lunch mostly because it causes them to have to adapt and find something else to drink or just simply drink water which sometimes people aren't in the mood to do. Sometimes people want variety and when they see their favorite drink run out they get upset. That's why we need a true solution to fixing the drink machine.

Problem Importance:

This problem is important to solve because it would allow people to enjoy their drinks, teach people how important it is to not waste things, and keep people from being upset. Many people don't like the situation with the drink machine and being able to solve this could keep many people at our school more satisfied at lunch because they know that instead of water they will actually get the drink they requested from the machine. Our goal is to create something that can either use the wasted syrup to create more or make it to where the drink machine can hold more syrup everyday,